

Task 5.1

Task 1.1: Superkey and Candidate Key Analysis

Relation A: Employee

1. Superkeys:

- {EmpID}
- {SSN}
- {Email}
- {EmpID, Name}
- {SSN, Department}
- {Email, Salary}

2. Candidate Keys:

EmpID, SSN, Email.

3. Primary Key:

EmpID, because it is unique, simple, and specifically identifies each employee.

4. Phone:

The sample shows different phone numbers, but Phone is not specified as unique. Therefore, two employees may have the same phone number.

Relation B: Course Registration

1. Primary Key:

(StudentID, CourseCode, Section, Semester, Year)

2. Explanation:

StudentID identifies the student, CourseCode the course, Section the section, and Semester + Year identify when the course was taken. All are needed to uniquely identify a registration.

3. Additional Candidate Keys:

No additional candidate keys are specified by the given business rules.

Task 1.2: Foreign Key Design

Foreign Keys:

- Student.AdvisorID → Professor.ProfID
- Course.DepartmentCode → Department.DeptCode
- Department.ChairID → Professor.ProfID

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- Enrollment.StudentID → Student.StudentID
- Enrollment.CourseID → Course.CourseID

Relationships:

- Professor 1:N Student
- Department 1:N Course
- Professor 1:1 Department
- Student 1:N Enrollment
- Course 1:N Enrollment

Student and Course have an M:N relationship through Enrollment.

Task 2.1.

Entities: Patient, Doctor, Department, Appointment, Prescription, Room.

Patient, Doctor, Department, Appointment and Prescription are strong entities. Room is a weak entity because RoomNumber is unique only inside a Department.

Special attributes:

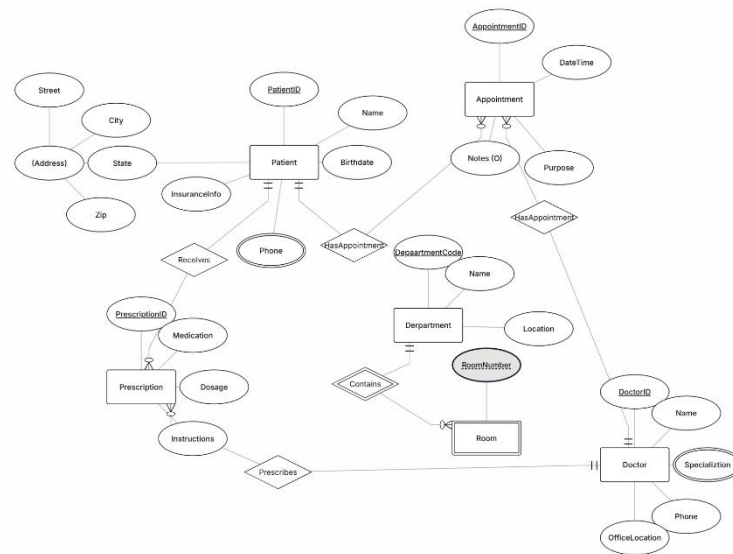
- Patient Address is composite: Street, City, State, Zip.
- Patient Phone is multivalued.
- Doctor Specialization is multivalued.

Main relationships:

- Patient 1:N Appointment
- Doctor 1:N Appointment
- Patient 1:N Prescription
- Doctor 1:N Prescription
- Department 1:N Room

Primary keys are PatientID, DoctorID, DepartmentCode, AppointmentID and PrescriptionID. Room uses Department + RoomNumber for identification.

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Task 2.2

The main entities are Customer, Order, Product, Category, Vendor, OrderItem, Review, Inventory and Address.

Weak Entity: OrderItem is a weak entity because it exists only as a part of an Order.

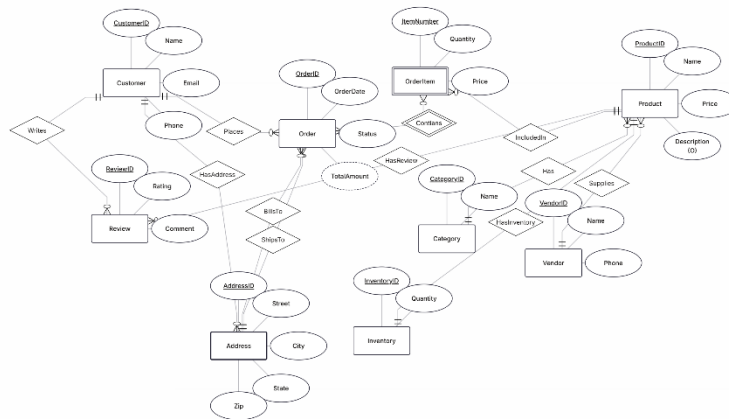
Main relationships:

- Customer 1:N Order
- Order 1:N OrderItem
- Product 1:N OrderItem
- Category 1:N Product
- Vendor 1:N Product
- Customer 1:N Review
- Product 1:N Review
- Product 1:1 Inventory

Order and Product have an M:N relationship through OrderItem. Quantity and Price are stored in OrderItem.

Shipping and billing addresses are stored separately, so they can be different.

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Task 4.1: Denormalized Table Analysis

StudentProject(StudentID, StudentName, StudentMajor, ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, Role, HoursWorked, StartDate, EndDate)

1. Functional Dependencies

$\text{StudentID} \rightarrow \text{StudentName}, \text{StudentMajor}$

$\text{ProjectID} \rightarrow \text{ProjectTitle}, \text{ProjectType}, \text{SupervisorID}, \text{StartDate}, \text{EndDate}$

$\text{SupervisorID} \rightarrow \text{SupervisorName}, \text{SupervisorDept}$

$(\text{StudentID}, \text{ProjectID}) \rightarrow \text{Role}, \text{HoursWorked}$

2. Problems

Redundancy: Student, project, and supervisor information is repeated for every student-project record.

Update anomaly: Changing a supervisor's department requires changing multiple rows.

Insert anomaly: A new project cannot easily be stored before a student joins it.

Delete anomaly: Deleting the last student from a project may also delete information about the project.

3. 1NF

The table satisfies 1NF if every attribute contains a single atomic value.

4. 2NF

Primary Key: (StudentID, ProjectID)

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Partial dependencies exist:

StudentID $\rightarrow$ StudentName, StudentMajor

ProjectID $\rightarrow$ ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, StartDate, EndDate

2NF decomposition:

Student(StudentID, StudentName, StudentMajor)

Project(ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, StartDate, EndDate)

StudentProject(StudentID, ProjectID, Role, HoursWorked)

5. 3NF

There is a transitive dependency:

ProjectID $\rightarrow$ SupervisorID $\rightarrow$ SupervisorName, SupervisorDept

Final 3NF decomposition:

Student(StudentID PK, StudentName, StudentMajor)

Supervisor(SupervisorID PK, SupervisorName, SupervisorDept)

Project(ProjectID PK, ProjectTitle, ProjectType, SupervisorID FK, StartDate, EndDate)

StudentProject(StudentID PK/FK, ProjectID PK/FK, Role, HoursWorked)

Task 4.2: Advanced Normalization

CourseSchedule(StudentID, StudentMajor, CourseID, CourseName, InstructorID, InstructorName, TimeSlot, Room, Building)

1. Primary Key

Primary Key: (StudentID, CourseID)

This combination identifies a student's enrollment in a course section.

2. Functional Dependencies

StudentID $\rightarrow$ StudentMajor

CourseID $\rightarrow$ CourseName, InstructorID, TimeSlot, Room

InstructorID $\rightarrow$ InstructorName

Room $\rightarrow$ Building

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(StudentID, CourseID) → all attributes

3. BCNF Check

The table is **not in BCNF** because some determinants are not candidate keys.

For example:

StudentID → StudentMajor

CourseID → CourseName, InstructorID, TimeSlot, Room

InstructorID → InstructorName

Room → Building

4. BCNF Decomposition

The table can be decomposed into:

Student(StudentID PK, StudentMajor)

Course(CourseID PK, CourseName)

Instructor(InstructorID PK, InstructorName)

Room(Room PK, Building)

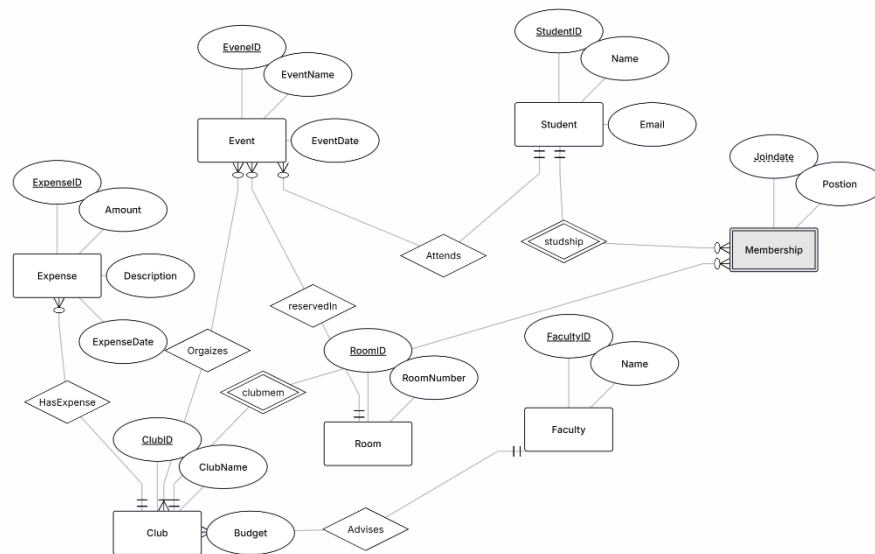
CourseSection(CourseID PK/FK, InstructorID FK, TimeSlot, Room FK)

Enrollment(StudentID PK/FK, CourseID PK/FK)

5. Loss of Information

The decomposition is designed to be lossless because the tables are connected using primary and foreign keys. The original information can be reconstructed by joining the decomposed tables.

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Task 5.1: University Clubs System

1. ER Diagram

The ER diagram includes Student, Club, Membership, Event, Faculty, Room, and Expense entities.

2. Relational Schema

Student(StudentID PK, Name, Email)

Club(ClubID PK, ClubName, Budget, FacultyID FK)

Membership(StudentID PK/FK, ClubID PK/FK, JoinDate, Position)

Event(EventID PK, EventName, EventDate, ClubID FK, RoomID FK)

Faculty(FacultyID PK, Name)

Room(RoomID PK, RoomNumber)

Expense(ExpenseID PK, Amount, Description, ExpenseDate, ClubID FK)

Attendance(StudentID PK/FK, EventID PK/FK)

3. Design Decision

Membership is used between Student and Club because students can join multiple clubs and each club can have multiple students. Position is stored in Membership to track officer roles such as President, Treasurer, and Secretary.

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4. Example Queries

1. Find all students who are members of a specific club.
2. List all events with their room reservations.
3. Find all expenses for a specific club.